

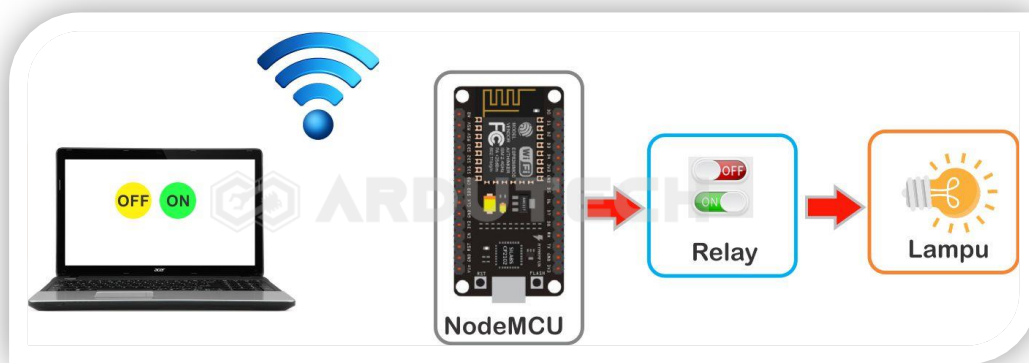
Kontrol 4 Lampu via WiFi

Deskripsi

Kontrol On – Off lampu secara *wireless* melalui WiFi. Sebagai controllernya NodeMCU ESP8266 yang dihubungkan dengan modul relay 4 channel.

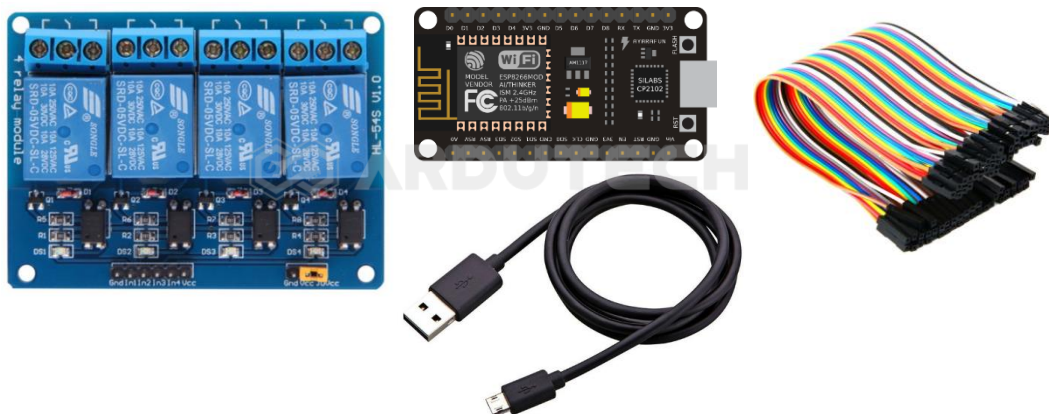
Cara Kerja

NodeMCU ESP8266 kita set sebagai server kemudian dihubungkan dengan jaringan WiFi (satu jaringan dengan device/perangkat yang dipakai untuk mengontrolnya). Dengan menekan tombol On – Off dari tampilan program kita dapat mengontrol lampu secara wireless.



Kebutuhan Bahan

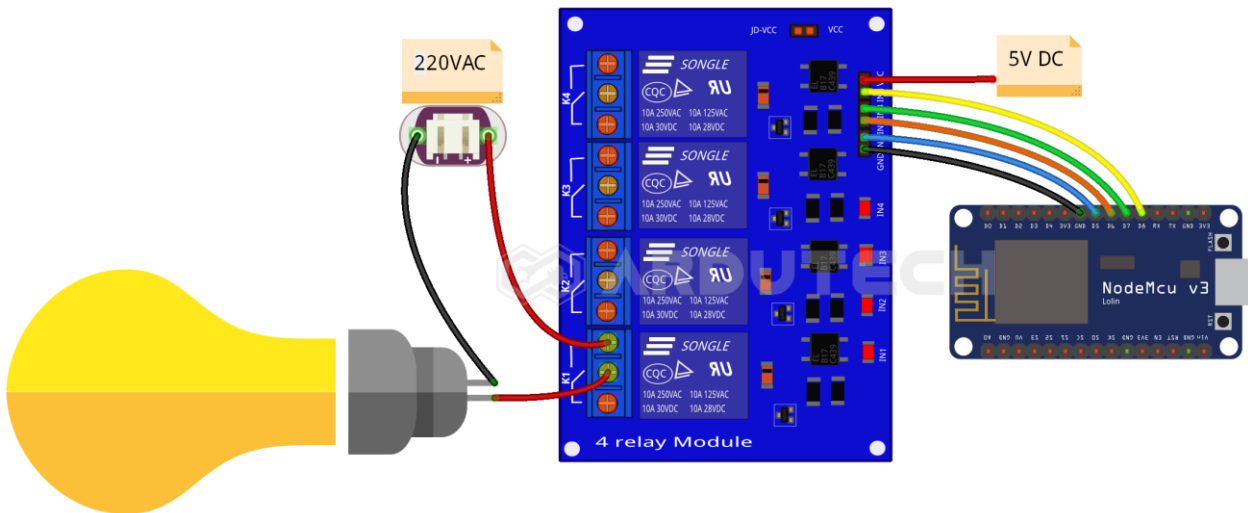
- NodeMCU ESP8266 V3
- Modul Relay 5V 4 Channel
- Kabel micro USB
- Kabel konektor



Kebutuhan Software

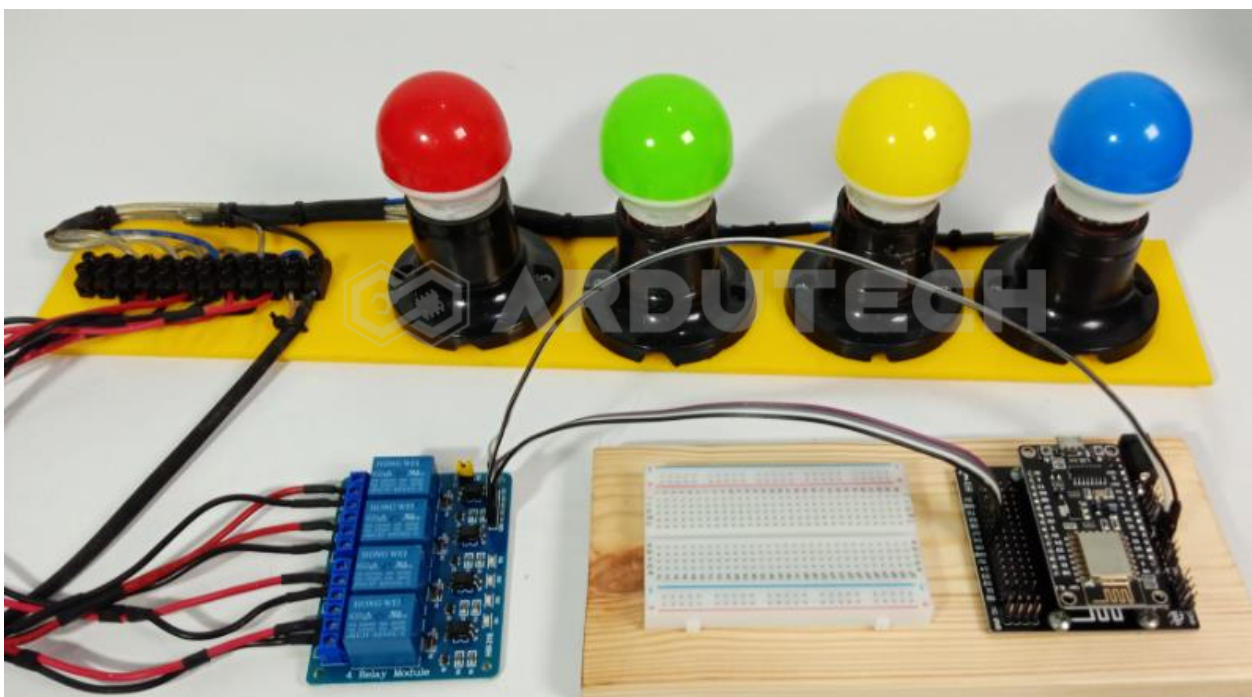
- Arduino IDE

Rangkaian/Skematik



Pada rangkaian hanya ada 1 lampu AC, untuk 3 lampu AC yang lain anda tinggal menambahkan sama dengan lampu 1.

NodeMCU V3	4 Ch Relay Module
D5	IN1
D6	IN2
D7	IN3
D8	IN4
GND	GND



Program/Source Code

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```

/*****

* Project Kontrol 4 Lampu via WiFi
* Board   : NodeMCU ESP8266 V3
* Input   : -
* Output  : Relay 4 Channels
* 99 Proyek IoT
* www.ardutech.com

*****/

#include <ESP8266WiFi.h>

//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
const char* ssid = "ArdutechWiFi"; //Nama Hotspot
const char* password = "12345678"; //Password

#define Relay1 D5
#define Relay2 D6
#define Relay3 D7
#define Relay4 D8

WiFiServer server(80);

//=====

void setup() {
    Serial.begin(9600);
    delay(10);
    pinMode(Relay1, OUTPUT);
    pinMode(Relay2, OUTPUT);
    pinMode(Relay3, OUTPUT);
    pinMode(Relay4, OUTPUT);
    digitalWrite(Relay1, LOW);
    digitalWrite(Relay2, LOW);
    digitalWrite(Relay3, LOW);

```

```
digitalWrite(Relay4, LOW);
Serial.println();
Serial.println();
Serial.print("Connecting to ");
Serial.println(ssid);

WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("");
Serial.println("WiFi connected");

// Start the server
server.begin();
Serial.println("Server started");
Serial.print("Use this URL to connect: ");
Serial.print("http://");
Serial.print(WiFi.localIP());
Serial.println("/");
}
//=====
void loop() {
    WiFiClient client = server.available();
    if (!client) {
        return;
    }
    Serial.println("new client");
    while(!client.available()){
        delay(1);
    }
}
```

```
}  
String request = client.readStringUntil('\r');  
Serial.println(request);  
client.flush();  
if (request.indexOf("/relay1on") > 0) {  
    digitalWrite(Relay1, HIGH);  
}  
if (request.indexOf("/relay1off") >0) {  
    digitalWrite(Relay1, LOW);  
}  
  
    if (request.indexOf("/relay2on") > 0) {  
        digitalWrite(Relay2, HIGH);  
    }  
    if (request.indexOf("/relay2off") >0) {  
        digitalWrite(Relay2, LOW);  
    }  
  
        if (request.indexOf("/relay3on") >0) {  
            digitalWrite(Relay3, HIGH);  
        }  
        if (request.indexOf("/relay3off") > 0) {  
            digitalWrite(Relay3, LOW);  
        }  
  
            if (request.indexOf("/relay4on") > 0) {  
                digitalWrite(Relay4, HIGH);  
            }  
        }
```

```
if (request.indexOf("/relay4off") > 0) {
    digitalWrite(Relay4, LOW);
}

client.println("HTTP/1.1 200 OK");
client.println("Content-Type: text/html");
client.println(""); // do not forget this one
client.println("<!DOCTYPE HTML>");
client.println("<html>");
client.println("<head>");
client.println("<meta name='apple-mobile-web-app-capable' content='yes'
/>");
client.println("<meta name='apple-mobile-web-app-status-bar-style'
content='black-translucent' />");
client.println("</head>");
client.println("<body bgcolor = \"#f7e6ec\">");
client.println("<hr/><hr>");
client.println("<h3><center> 99 Project IoT : Kontrol 4 Lampu
</center></h3>");
client.println("<h4><center> www.ardutech.com  </center></h4>");
client.println("<hr/><hr>");
client.println("<br><br>");
//client.println("<br><br>");
client.println("<center>");
client.println("Lampu 1");
client.println("<a href=\"/relay1on\"><button>Turn On
</button></a>");
client.println("<a href=\"/relay1off\"><button>Turn Off
</button></a><br />");
client.println("</center>");
client.println("<br><br>");
```

```

client.println("<center>");
client.println("Lampu 2");
client.println("<a href=\""/relay2on\""/><button>Turn On
</button></a>");
client.println("<a href=\""/relay2off\""/><button>Turn Off
</button></a><br />");
client.println("</center>");
client.println("<br><br>");
client.println("<center>");
client.println("Lampu 3");
client.println("<a href=\""/relay3on\""/><button>Turn On
</button></a>");
client.println("<a href=\""/relay3off\""/><button>Turn Off
</button></a><br />");
client.println("</center>");
client.println("<br><br>");
client.println("<center>");
client.println("Lampu 4");
client.println("<a href=\""/relay4on\""/><button>Turn On
</button></a>");
client.println("<a href=\""/relay4off\""/><button>Turn Off
</button></a><br />");
client.println("</center>");
client.println("<br><br>");
client.println("<center>");
client.println("<table border=\"5\">");
client.println("<tr>");
//=====
if (digitalRead(Relay1))
{
    client.print("<td>Lampu 1 = ON</td>");
}
else

```

```
{
    client.print("<td>Lampu 1 = OFF</td>");
}
client.println("<br />");
//=====
if (digitalRead(Relay2))
{
    client.print("<td>Lampu 2 = ON</td>");
}
else
{
    client.print("<td>Lampu 2 = OFF</td>");
}
client.println("</tr>");
//=====
client.println("<tr>");

if (digitalRead(Relay3))
{
    client.print("<td>Lampu 3 = ON</td>");
}
else
{
    client.print("<td>Lampu 3 = OFF</td>");
}
if (digitalRead(Relay4))
{
    client.print("<td>Lampu 4 = ON</td>");
}
else
{
```



```
        client.print("<td>Lampu 4 = OFF</td>");  
    }  
    client.println("</tr>");  
    client.println("</table>");  
    client.println("</center>");  
    client.println("</html>");  
    delay(1);  
    Serial.println("Client disconnected");  
    Serial.println("");  
}
```

PERHATIKAN !

Ganti/sesuaikan variabel berikut :

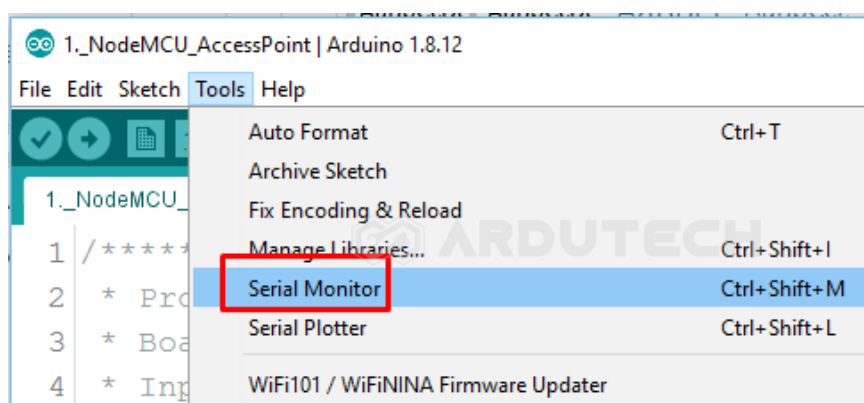
- Nama jaringan WiFi/hotspot : **ssid**
- Password jaringan WiFi/hotspot : **password**

Sesuaikan nama ssid (hotspot) dan password-nya dengan hotspot anda.

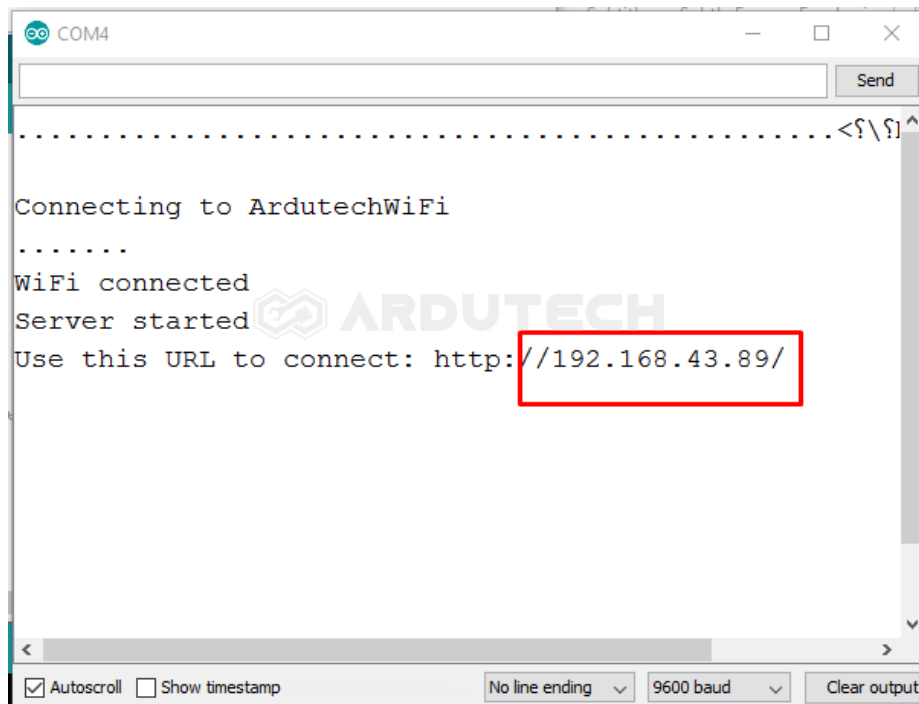
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

Jalannya Alat

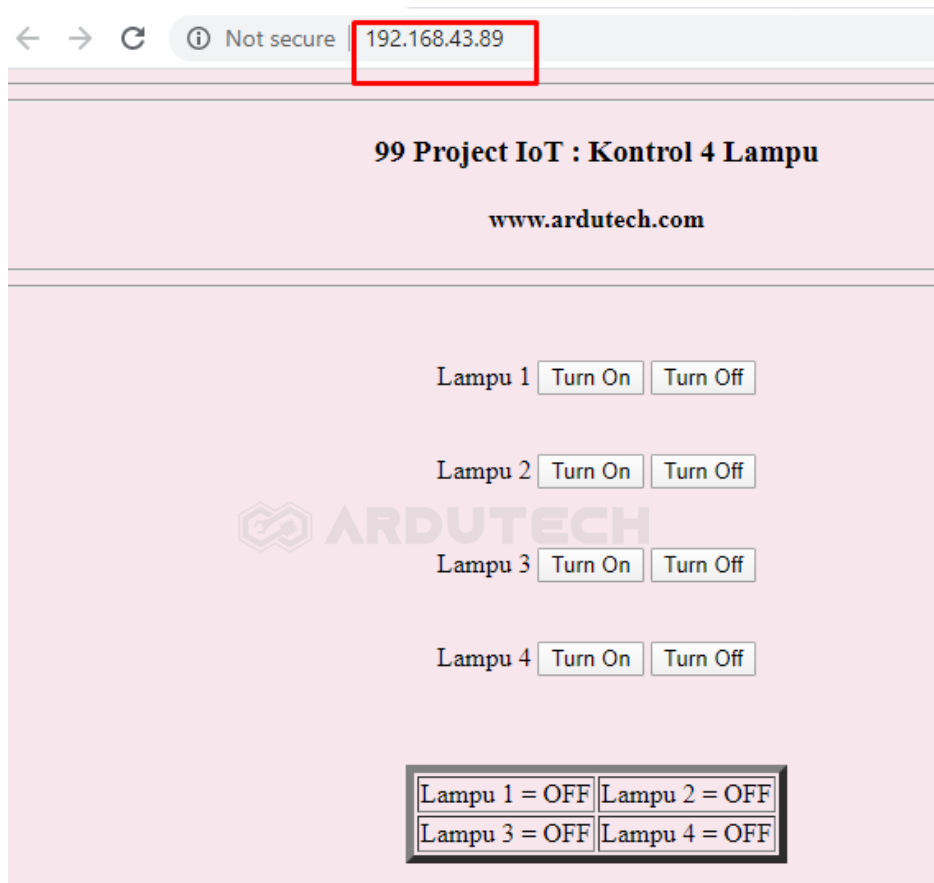
Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools → Serial Monitor** kemudian seting baudrate pada nilai 9600.



Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :



Stelah diketahui IP address-nya silakan ketikkan alamat tersebut di browser anda sehingga tampil :



Tekan tombol lampu 1 **“Turn On”** sehingga relay 1 **“ON”** dan menyalakan lampu 1. Coba juga untuk tombol yang lain.



Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – *“Sahabat Inovasi Anda”*